AI APPLICATION DEVELOPMENT INTERN · TAKE-HOME

# Best Buy PC Price Tracker

Comparing prices only when the machines are truly the same

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# About me

EDUCATION

**M.S. Information Security**
Taipei Tech, 2026–

**B.S. Computer Science**
National Taitung University
GPA 3.74 / 4.0

FOUNDATION

**CPE top 5.3%**
competitive programming exam

**Python, FastAPI**
pytest, CI, LLM APIs

PROJECTS

**LLM code grader**
CodeLlama 13B on a school RTX 5090, for courses

**PC price tracker**
today's talk

# The task: track prices of comparable PCs

A price chart is only useful if every line compares the same machine.

RETAILER

**Best Buy, US online**

PRODUCTS

**6 Windows PCs, 4 brands**

DATA SO FAR

**24 prices, 11–13 Sept**

# Equivalence is code, not judgement

```
intel-ultra5-226v | 16gb | 512gb | windows-11-pro | laptop | clamshell
```

| Key | Rule | Question it answers |
|---|---|---|
| group_key | All six parts identical | Is it the same machine? |
| tier_key | Processor reduced to its class | Would a buyer cross-shop? |

Result: 6 products → 5 exact groups, 3 tiers. **The largest tier compares 4 brands.**

# When unsure, exclude. Never guess.

WHAT HAPPENED

Two titles said only "Intel Core Ultra 5". The parser refused to guess a model.

WHAT I FOUND

Best Buy's own field said "N150". The descriptions said 225U and 236V.

## A wrong merge looks correct. A missing line is visible.

# Four sources, tried in order

| # | Source | What happened |
|---|---|---|
| 1 | Best Buy Products API | Built and tested; key not available |
| 2 | RapidAPI aggregator | 9 prices on 11 Sept; free quota then used up |
| 3 | Page extraction (Playwright) | Blocked by bot protection |
| 4 | Manual entry + screenshot | 15 prices, 11–13 Sept |

Every price records its source. **If every source fails, no price is written**: the run raises a flag instead of a guess.

# The model is constrained, not trusted

### Spec extraction

Reads specs that only appear in prose. Output goes back through the rules. If rules disagree, a flag is raised.

### Tool calling

The assistant picks from 5 query functions. Arguments are schema-checked. No query, no "verified" label.

### Retrieval (BM25)

No relevant passage: the model is not called. Citations are checked against real passage ids.

LIVE DEMO · ABOUT 5 MINUTES

# From a price chart to the audit trail

**1**

### Price chart

Same tier vs exact match

**2**

### Review queue

Flagged for a person, never auto-fixed

**3**

### Ask a price

Answer plus the tool calls behind it

**4**

### It declines

No matching passage, no model call

**5**

### Commit history

Every snapshot is a commit

Every number on screen can be traced to a query, a passage or a commit.

Backup: a recording of the same flow, if the network fails

# Measure it before believing it

RULES ONLY

**74.1%**

Title only; misses specs it never sees

LLM ONLY

**96.3%**

Title and description

RULES + LLM, RECONCILED

**98.1%**

The two miss different things

Field accuracy on **9 hand-checked cases**, 2 of them traps where the answer is "not stated". **0 hallucinations** in all three.

```
python tools/run_eval.py --llm → scores rules, model and hybrid
```

# Quality checks on every push

**147**
pytest tests

**78%**
line coverage, gate in CI

**11**
type errors caught by mypy

**2**
separate workflows: tests and collection

A failing test never stops price collection, and a collection outage never looks like a broken build. **A grouping guard checks the real product list in CI.**

# How I used AI to build it

AI DID

Most of the code: collection pipeline, parser, validation rules, tests, CI.

I DID

Direction and trade-offs: kept pushing for automated collection, saw that v1 had no AI, set the scope, verified with tools.

## Code that passes its tests is not the same as code that is correct.

# Limitations and next steps

| Limitation | Next step |
| --- | --- |
| API free quota used up; schedule paused since 14 Sept | Paid quota or the official key, then resume |
| Only 24 prices, 11–13 Sept | Keep collecting; set alert thresholds from real data |
| Prices are second-hand | Switch to the official API once a key is available |
| Screen size is not in the key | Add it when there are enough products |
| BM25 can't tell "Wi-Fi 7" from "Wi-Fi 6E" | Swap in embeddings; only one class changes |

# What I would bring to the team

01 Build working AI tools fast, with AI assistance

02 Measure LLM output before trusting it

03 A security mindset: validate inputs, constrain the model

**Thank you. Questions?**